



QRT Inc.

Reliability Test Report

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COMPANY : UBIVELOX Inc.

PRODUCT : UONE-A03

PACKAGE : 20 QFN

PURPOSE : Qualification

ISSUED DATE : 2023-03-02

REFERENCE No. : 2211-19075

Prepared By : YoonGa Young

윤가영

Approved By : Youngran Kim

조

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1 Qualification Plan and Results

The purpose of these tests is evaluation (or monitoring) of **UONE-A03**.

Test conditions are specified in customers in-house test plan and all test procedures comply with **JEDEC** standards.

Obligation to perform exact procedure to reference documents is QRT Inc. responsibility only, but establishing failure criteria and judgment of pass/fail is customer's responsibility.

Test Items	Test Condition	Duration	Sample Size	Failed Unit ¹⁾	Test Method ²⁾
Precondition	BAKE : 125 (+5, -0) °C SOAK : 30 °C ± 2 °C, 60 % ± 3 % R.H. REFLOW : ≥ 260 °C	24 h 192 h 3 Cycles	120	0	JESD22-A113I:2020 (PRECONDITIONING OF NONHERMETIC SURFACE MOUNT DEVICES PRIOR TO RELIABILITY TESTING)
High Temperature Operating Life	125 °C ± 5 °C V _O = 5.5 V, V _{IH} = 3.6 V	1 000 h	76	0	JESD22-A108F:2017 (TEMPERATURE, BIAS, AND OPERATING LIFE)
High Temperature Storage	150 (+10, -0) °C	1 000 h	45	0	JESD22-A103E.01:2021 (HIGH TEMPERATURE STORAGE LIFE)
^ Unbiased-Highly Accelerated Stress Test	130 °C ± 2 °C, 85 % ± 5 % R.H. / 230 kPa	96 h	50	0	JESD22-A118B.01:2021 (ACCELERATED MOISTURE RESISTANCE -UNBIASED HAST)
^ Temperature Cycling	-65 (+0, -10) °C ~ 150 (+10, -0) °C	1 000 Cycles	45	0	JESD22-A104F:2020 (TEMPERATURE CYCLING)
Electrostatic Discharge (ESD) Human Body Model	± 2 000 V	-	9	0	JS-001-2017 (HUMAN BODY MODEL (HBM) ELECTROSTATIC DISCHARGE TEST)
Electrostatic Discharge (ESD) Charged Device Model	± 200 V	-	9	0	JS-002-2018 (FIELD-INDUCED CHARGEDDEVICE MODEL TEST METHOD FOR ELECTROSTATIC DISCHARGE WITHSTAND THRESHOLDS OF MICRO ELECTRONIC COMPONENTS)
Latch-Up	I-Test : ± 100 mA (Test Temp : Room) V _{supply} over test : + 8.25 V (Test Temp : Room)	-	6	0	JESD78F:2022 (IC LATCH-UP TEST)

* This test report is not related to KOLAS accreditation.

Additional Requirement : N/A

Note :

1) Test results are based on the "Final Test Result" provided by customer.

2) Although the name of test item is same, reference documents can be JEDEC.

^ "Preconditioning" was applied before reliability test.

2 Test Vehicles

Manufacturer : UBIVELUX Inc.

Part Number : UONE-A03

Description : -

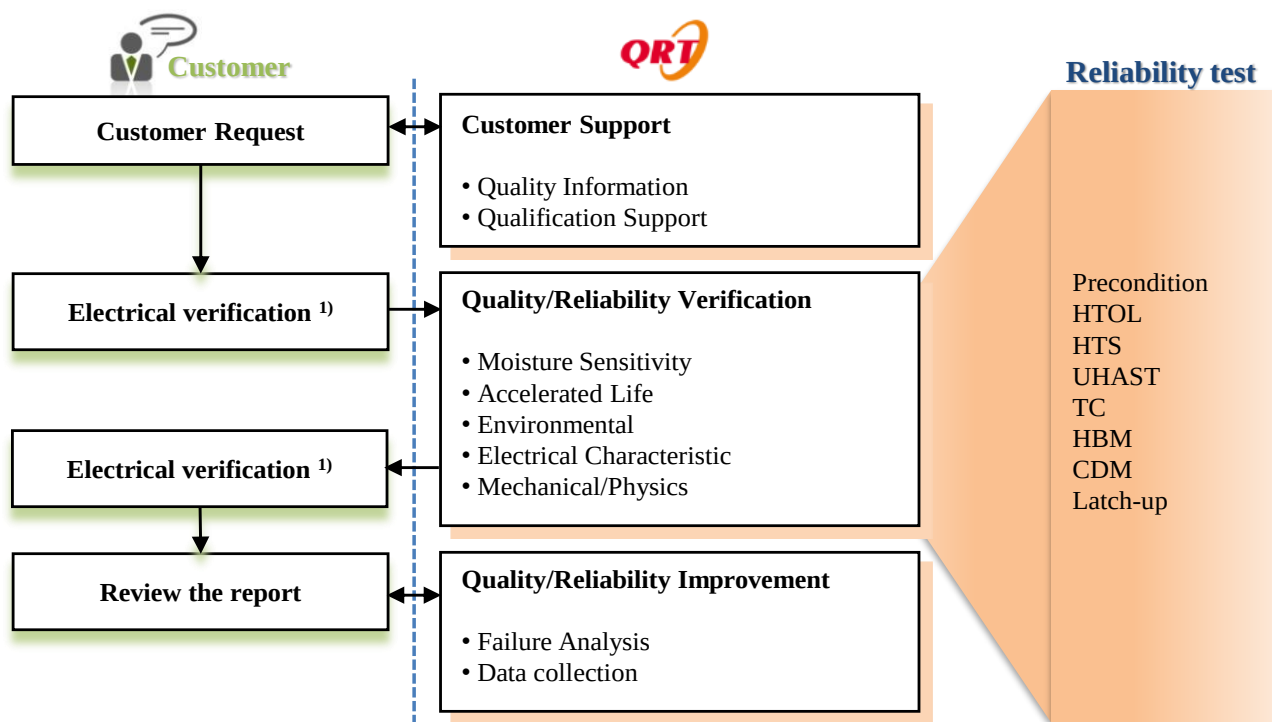
Package Type : 20 QFN

Ass'y Site : -

Fab Site : -

3 Test Flow/Procedure

All reliability tests performed by QRT Inc. inc. comply with international standard. QRT inc. provide customer support featuring easy-understanding documents & technical information. And product monitoring activities to confirm quality and reliability level by sampling inspection and reliability test.



NOTE ¹⁾ Customer's Responsibilities only. Electrical Test can be omitted depending on customer's request

Figure 3-1. Reliability Test Flow

4 Preconditioning (including MSL)

Preconditioning of SMD packages is used to simulate the effects of board assembly on moisturized packages, prior to reliability testing. This allows reliability testing at the component level on as shippable products with a board assembly simulation. During preconditioning, test samples are subjected to temperature cycling (optional), dry bake, moisture soaking, solder reflow simulation, flux, rinse, dry, and electrical test before reliability testing.

4.1 Preconditioning Procedure

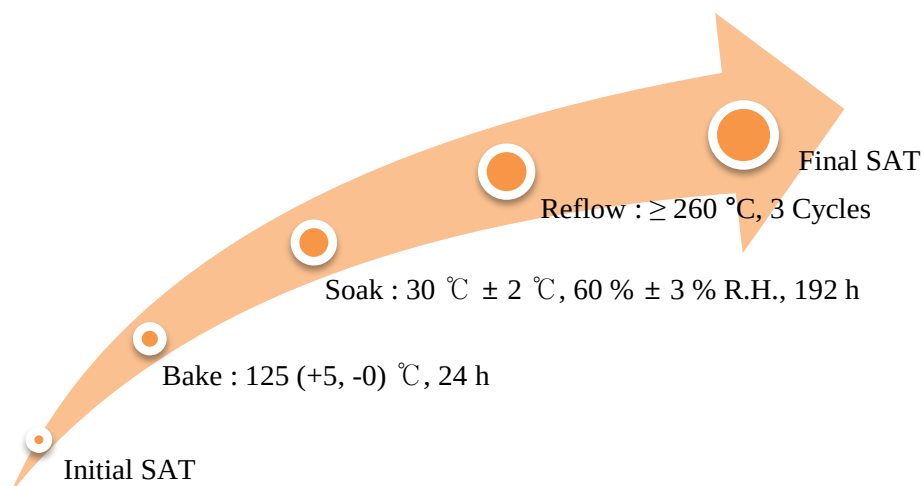


Figure 4-1. Preconditioning Flow

4.2 Reflow Profile

Reflow Profile was set using 3 representative components. All temperatures refer to the center of the package, measured on the package body surface that is facing up during assembly reflow.

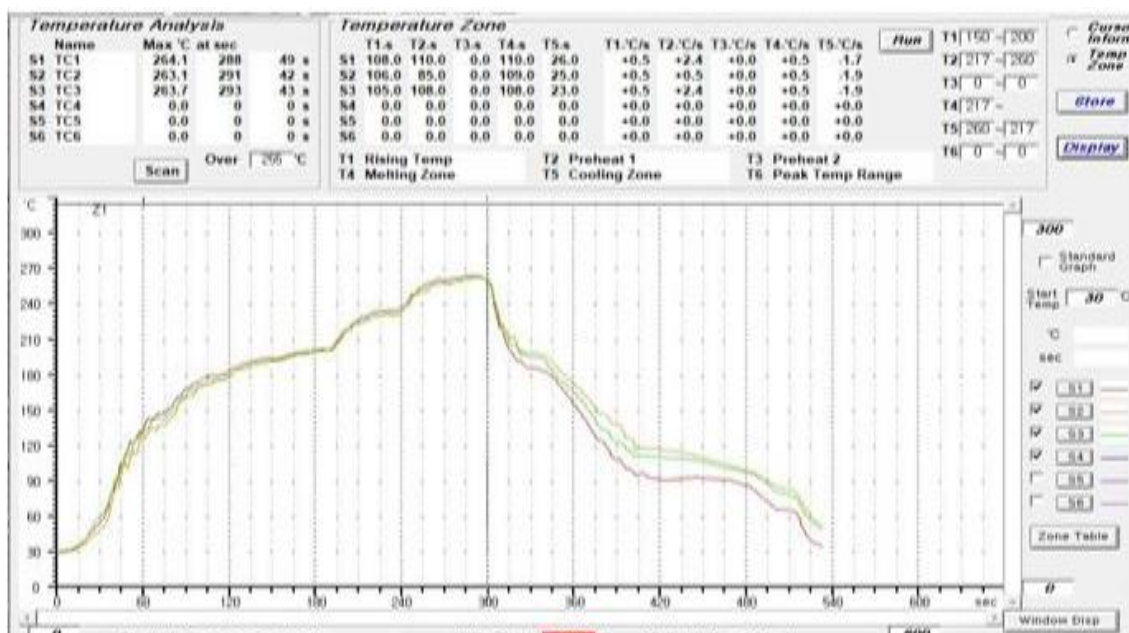


Figure 4-2. Reflow Profile

5 SAT Analysis

5.1 SAT Inspection

Sample	Defect List	Result				Comments	
		Yes	No	N/A	If Yes		
					Location		
Initial	Delamination		○		Die surface to molding		
					Mold to lead frame(paddle)		
					Die to adhesive tape		
					Die to substrate		
					Molding to substrate		
					Others:		
	Crack		○		Chip		
					Molding compound		
					Substrate		
					Others:		
	Others						
	After Reliability Test	Delamination		○		Die surface to molding	
						Mold to lead frame(paddle)	
Die to adhesive tape							
Die to substrate							
Molding to substrate							
Others:							
Crack			○		Chip		
					Molding compound		
					Substrate		
					Others:		
Others							
Comment : -							

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

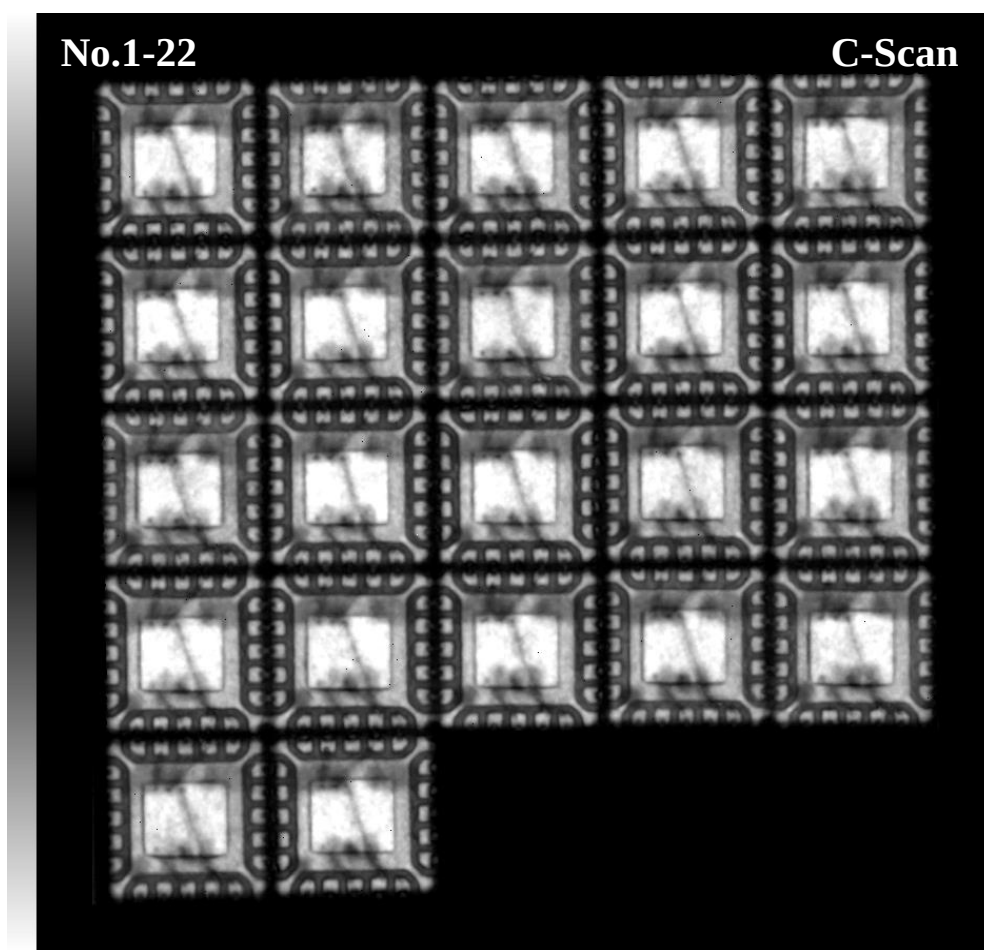


Figure.5-1 SAT images before Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

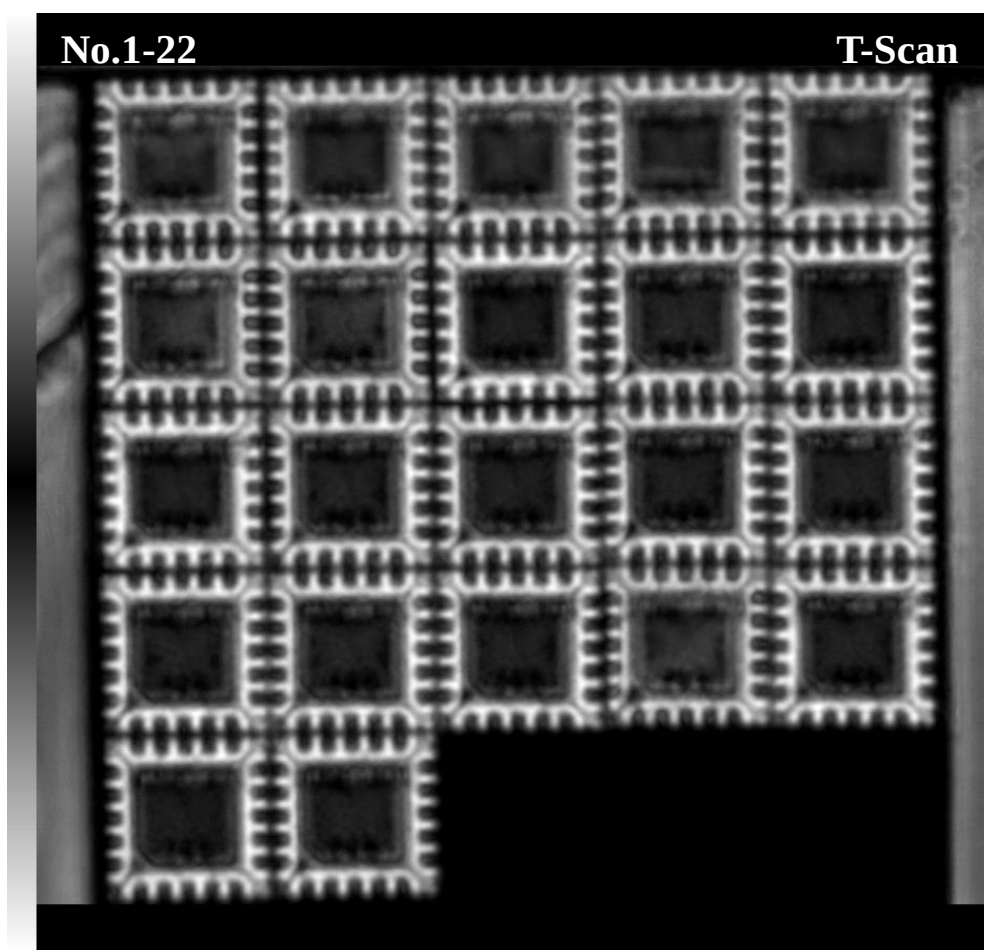


Figure.5-2 SAT images before Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

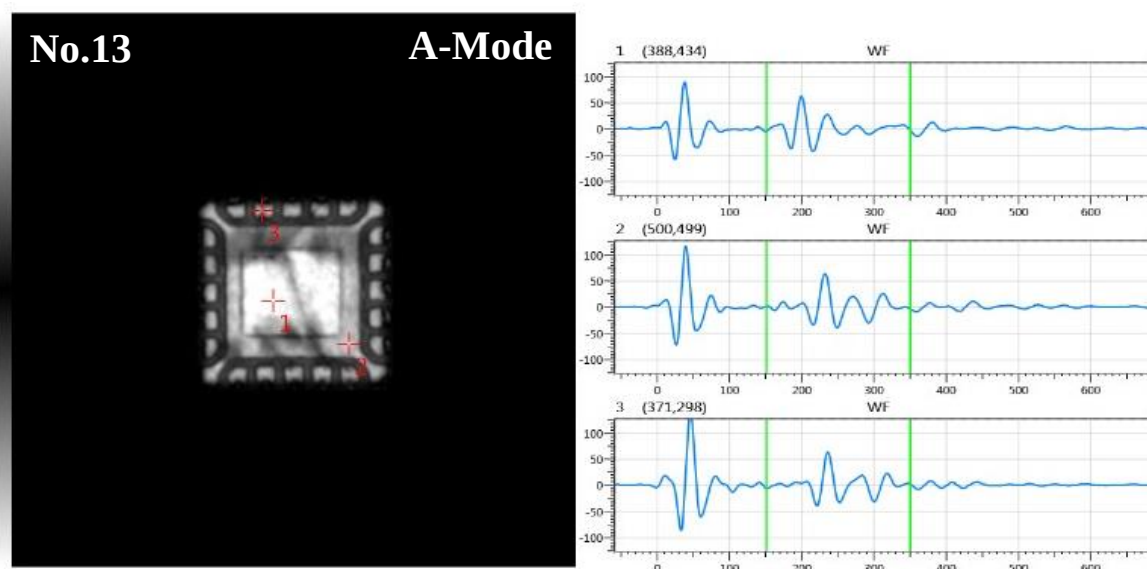


Figure.5-3 SAT Waveform Data before Precondition

5.1 SAT Inspection (cont'd)

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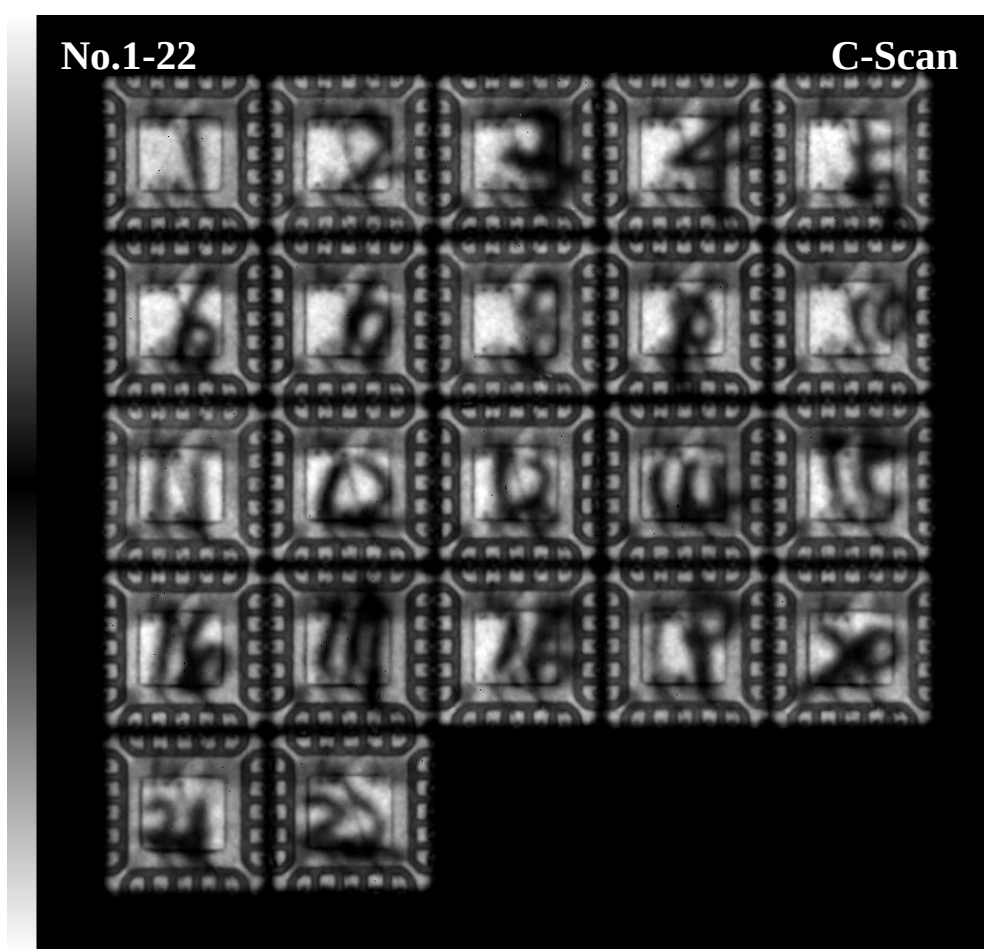


Figure.5-4 SAT images after Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object. It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

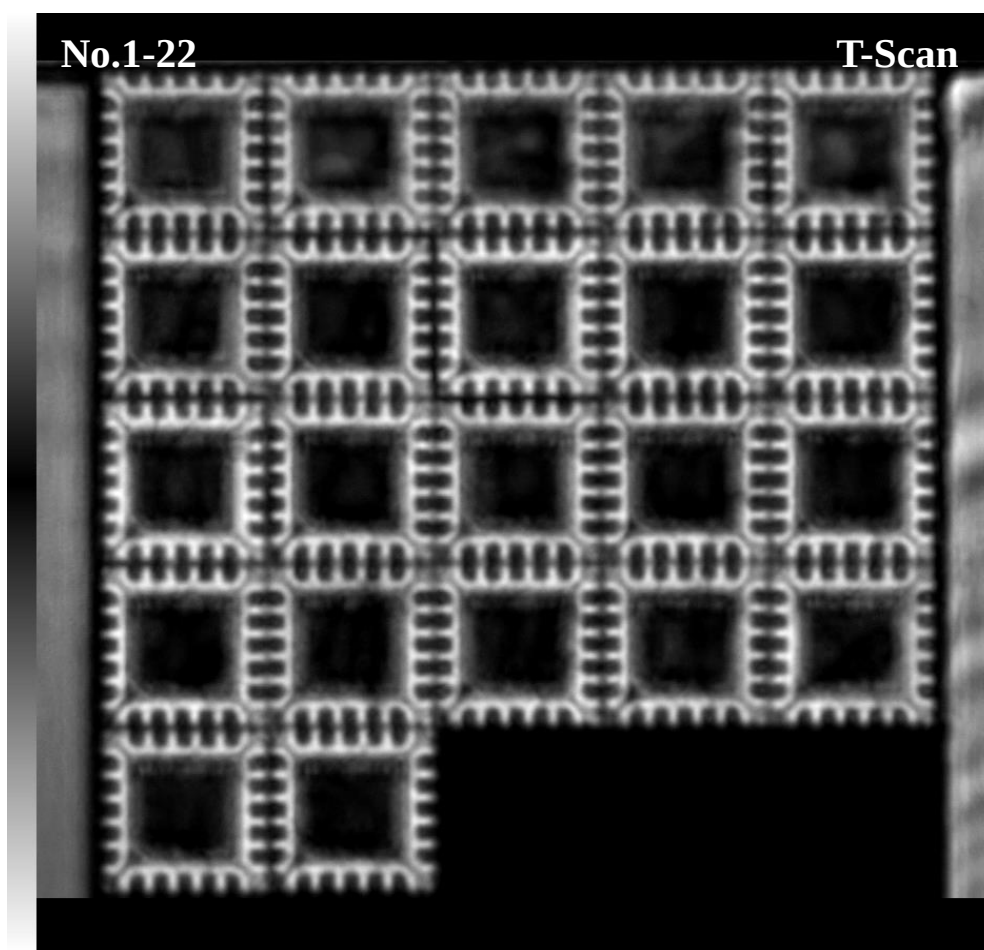


Figure.5-5 SAT images after Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

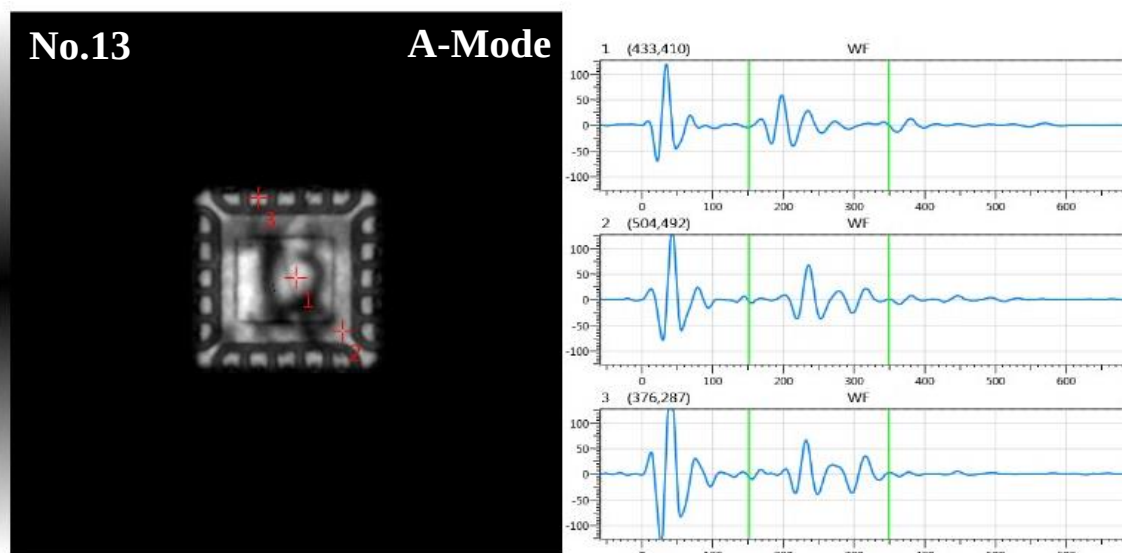


Figure.5-6 SAT Waveform Data after Precondition

6 Reliability Tests

6.1 High Temperature Operating Life (HTOL)

The High Temperature Operating Life (HTOL) or steady-state life test is performed to determine the reliability of devices under operation at high temperature conditions over an extended period of time. It consists of subjecting the parts to a specified bias or electrical stressing, for a specified amount of time, and at a specified high temperature.

Test condition

Chamber condition : $125\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$
Bias configuration : $V_0 = 5.5\text{ V}$, $V_{IH} = 3.6\text{ V}$
Duration : 1 000 h
Sample Size : 76 ea

6.2 High Temperature Storage (HTS)

This test is typically used to determine the effects of time and temperature, under storage conditions, for thermally activated failure mechanisms and time-to failure distributions of solid state electronic devices, including nonvolatile memory devices (data retention failure mechanisms). Thermally activated failure mechanisms are modeled using the Arrhenius Equation for acceleration. During the test, accelerated stress temperatures are used without electrical conditions applied.

Test Condition

Chamber condition : $150 (+10, -0)\text{ }^{\circ}\text{C}$
Duration : 1 000 h
Sample Size : 45 ea

6.3 Unbiased-Highly Accelerated Stress Test (UHAST)

This test is performed for the purpose of evaluating the reliability of non-hermetic packaged solid-state devices in humid environments. It is a highly accelerated test which employs temperature and humidity under non-condensing conditions to accelerate the penetration of moisture through the external protective material (encapsulant or seal) or along the interface between the external protective material and the metallic conductors which pass through it. Bias is not applied in this test to ensure the failure mechanisms potentially overshadowed by bias can be uncovered (e.g., galvanic corrosion).

Test Condition

Chamber condition : $130\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$, $85\text{ \%} \pm 5\text{ \% R.H.}$ / 230 kPa

Duration : 96 h

Sample Size : 50 ea

6.4 Temperature Cycling (TC)

This test is conducted to determine the ability of components and solder interconnects to withstand mechanical stresses induced by alternating high- and low-temperature extremes. Permanent changes in electrical and/or physical characteristics can result from these mechanical stresses.

Test Condition

Chamber condition : $-65\text{ }(^{-0}, +10)\text{ }^{\circ}\text{C} \sim 150\text{ }(^{-10}, +0)\text{ }^{\circ}\text{C}$

Duration : 1 000 Cycles

Sample Size : 45 ea

7. ESD Test Results

7.1 HBM Test Results

Pin Combinations Sets		No. of Zaps (Polarity)	Test Level	Sample Size	Test Result ¹⁾	Classification
Test Pin (Terminal A)	Ref. Ground (Terminal B)					
VSS, I/O	VPP	1 (±)	2 000 V	3 ea	Pass	2
VPP, I/O	VSS		2 000 V	3 ea		
Every I/O	All I/O s (except for test pin)		2 000 V	3 ea		
NOTE						
1) Functional test conducted by customer site - Reference Document : JS-001-2017						

7.2 CDM Test Results

Charge Method (Dielectric)	Discharge Pin	No. of Zaps (Polarity)	Test Level	Sample Size	Test Result ¹⁾	Classification
Field Induced (FR-4)	Every pin (except for N/C)	1 (±)	200 V	9 ea	Pass	C0b
NOTE 1) Functional test conducted by customer site - Reference Document : JS-002-2018						

8. Latch-up Test Results

Stress Type (Test Method) (Temperature)	Test Pin or Pin Group	Condition of untested input pins	(Polarity) Stress Level	Sample Size	Test Result		Immunity Level
					Latch-up Detection ¹⁾	F/T ²⁾	
Signal Pin Test (I-Test) (Room)	I/O	Maximum pin Operating Voltage & Minimum pin Operating Voltage	(±) 100 mA	3 ea	Pass	Pass	A
Supply Test (Room)	Power		(+) 8.25 V	3 ea	Pass	Pass	
NOTE							
1) Latch-up Detection Criteria							
a. If absolute Inom ≤ 25 mA, then absolute Inom + 10mA							
b. If absolute Inom > 25 mA, then absolute Inom × 1.4							
2) Functional test conducted by customer site							
- Reference Document : JESD78F:2022							

9. Descriptions of ESD (ElectroStatic Discharge)

Electrostatic Discharge, or ESD, is a single-event, rapid transfer of electrostatic charge between two objects, usually resulting when two objects at different potentials come into direct contact with each other. ESD can also occur when a high electrostatic field develops between two objects in close proximity. ESD is one of the major causes of device failures in the semiconductor industry.

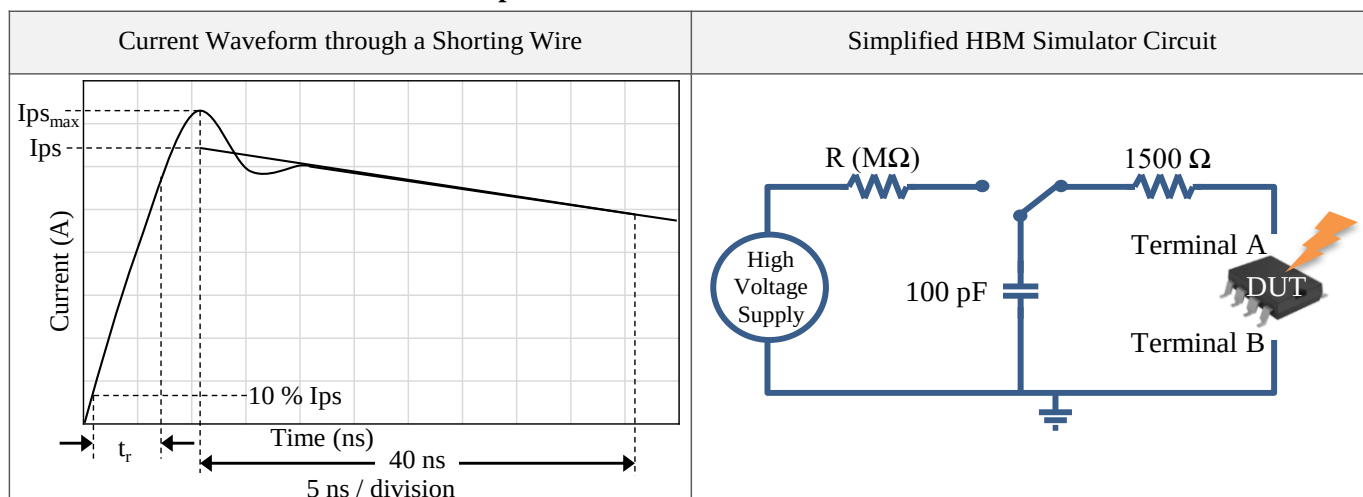
9.1 HBM (Human Body Model)

This test is classifying microcircuits according to their susceptibility to damage or degradation by exposure to a defined electrostatic Human Body Model (HBM) discharge (ESD).

9.1.1 HBM ESD Component Classification Levels

Classification	Voltage Range (V)
0A	< 125 V
0B	125 to < 250 V
1A	250 to < 500 V
1B	500 to < 1000 V
1C	1000 to < 2000 V
2	2000 to < 4000 V
3A	4000 to < 8000 V
3B	≥ 8000 V

9.1.2 HBM Current Waveform & Simplified HBM Simulator Circuit



9.1.3 HBM Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
MK2	Thermo Fisher Scientific	401208	2022.11.04 ~ 2023.11.04	SICT(KC01-018)
MK4. TE	Thermo Fisher Scientific	1806237	2022.11.04 ~ 2023.11.04	

9. Descriptions of ESD (ElectroStatic Discharge)

Electrostatic Discharge, or ESD, is a single-event, rapid transfer of electrostatic charge between two objects, usually resulting when two objects at different potentials come into direct contact with each other. ESD can also occur when a high electrostatic field develops between two objects in close proximity. ESD is one of the major causes of device failures in the semiconductor industry.

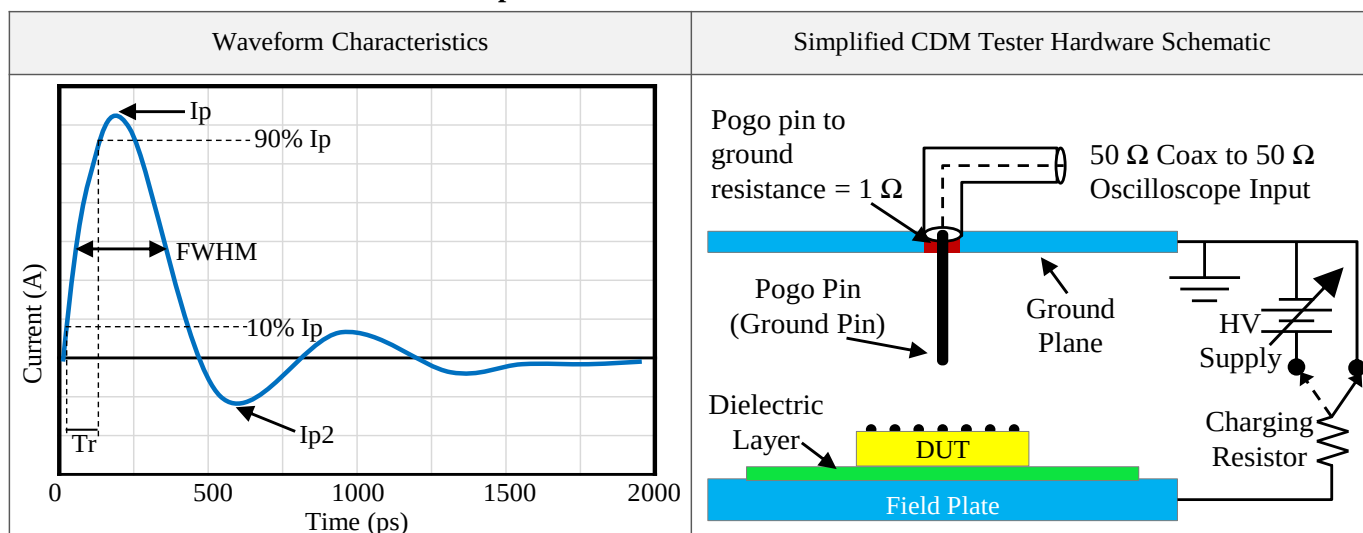
9.2 CDM (Charged Device Model)

This test is classifying microcircuits according to their susceptibility to damage or degradation by exposure to a defined electrostatic Charged Device Model (CDM) discharge (ESD).

9.2.1 CDM ESDS Device Classification Levels

Classification Level	Classification Test Condition (in Volts)
C0a	< 125
C0b	125 to < 250
C1	250 to < 500
C2a	500 to < 750
C2b	750 to < 1000
C3	≥ 1000

9.2.2 CDM Current Waveform & Simplified CDM Tester Hardware Schematic



9.2.3 CDM Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
Orion3	Thermo Fisher Scientific	1706181	2022.04.25 ~ 2023.04.25	SICT(KC01-018)
Orion3	Thermo Fisher Scientific	2006173	2022.04.25 ~ 2023.04.25	

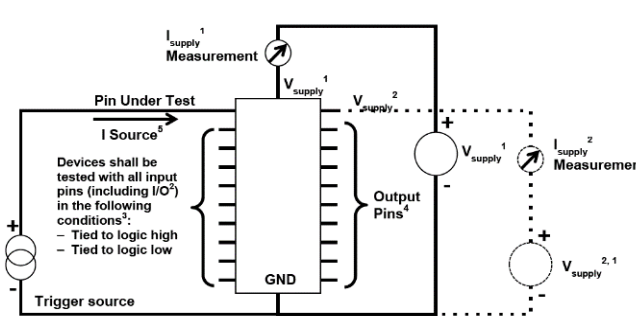
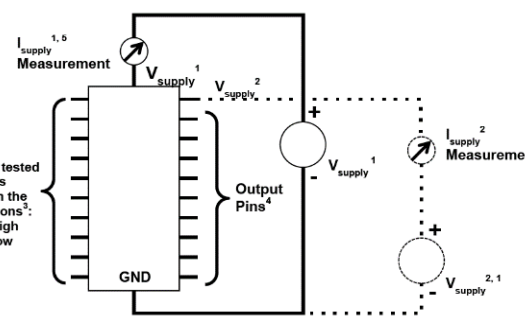
10. Descriptions of Latch-up Test

Latch-up pertains to a failure mechanism wherein a parasitic thyristor (such as a parasitic silicon controlled rectifier, or SCR) is inadvertently created within a circuit, causing a high amount of current to continuously flow through it once it is accidentally triggered or turned on. Depending on the circuits involved, the amount of current flow produced by this mechanism can be large enough to result in permanent destruction of the device due to electrical overstress (EOS).

10.1 Latch-up immunity levels

Immunity Level	Test Type	Magnitude of Trigger Force Current or Voltage
A	Positive Signal Pin Test ¹⁾	$\geq 100 \text{ mA}$
	Negative Signal Pin Test ¹⁾	The actual injected current may be less than 100 mA, if pin voltage preset limits are reached
	Supply Test	1.5 x VDD(Max.) or MSV, whichever is less The actual applied voltage may be less than 1.5 x V _{maxSUP} or MSV, if supply current limits are reached
B	If immunity level A cannot be achieved	
(1) In the Signal Pin Test, current injection is achieved either by current forcing with voltage compliance limit (I-Test) or by applying voltage with current compliance limit (E-Test).		

10.2 The equivalent circuit for Latch-up Test

Signal Pin Test(I-Test or E-Test)	Supply Test
 Devices shall be tested with all input pins (including I/O²) in the following conditions²: - Tied to logic high - Tied to logic low	 Devices shall be tested with all input pins (including I/O²) in the following conditions²: - Tied to logic high - Tied to logic low

10.3 Latch-up Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
MK2	Thermo Fisher Scientific	401208	2022.11.04 ~ 2023.11.04	SICT(KC01-018)
MK4. TE	Thermo Fisher Scientific	1806237	2022.11.04 ~ 2023.11.04	

11. Test Pin Group

11.1 ESD (HBM, CDM) Test Pin Group

Pin Group	Type	Pin(s)
VPP	POWER	14
VSS	GROUND	3
I/O	I/O	4,5,12
N/C	N/C	1,2,6-11,13,15-20

11.2 Latch-up Test Pin Group

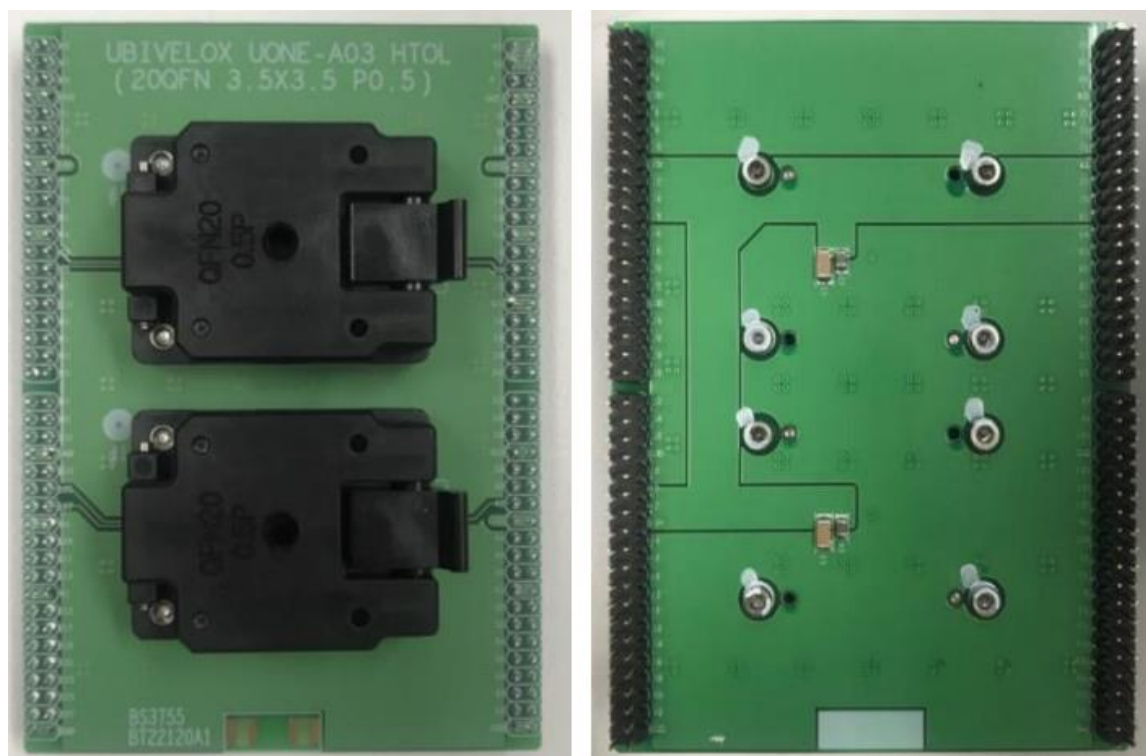
Pin Group	Type	Pin(s)
VPP_5.5	POWER	14
VSS	GROUND	3
High_5.5	I/O	4
I/O_5.5	I/O	5,12
N/C	N/C	1,2,6-11,13,15-20

APPENDIX A: HTOL Test Conditions

1. HTOL Test Configuration

Test Item : HTOL	Test condition : 125 °C
V0 : 5.5 V	VIH : 3.6 V
Vector Type : -	Cycle Time : 1000 ns
Timing Format Condition	
CLK : 1Mhz / CH83, CH105	

2. HTOL Test Board

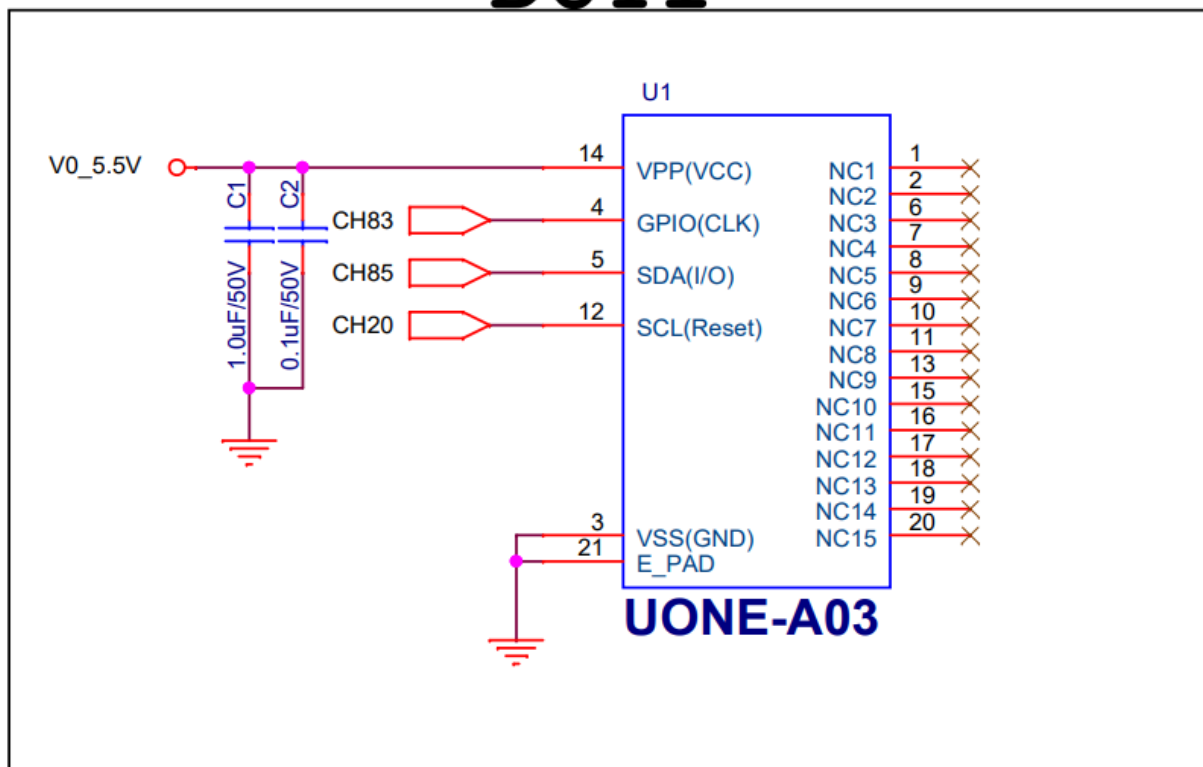


3. Board Information

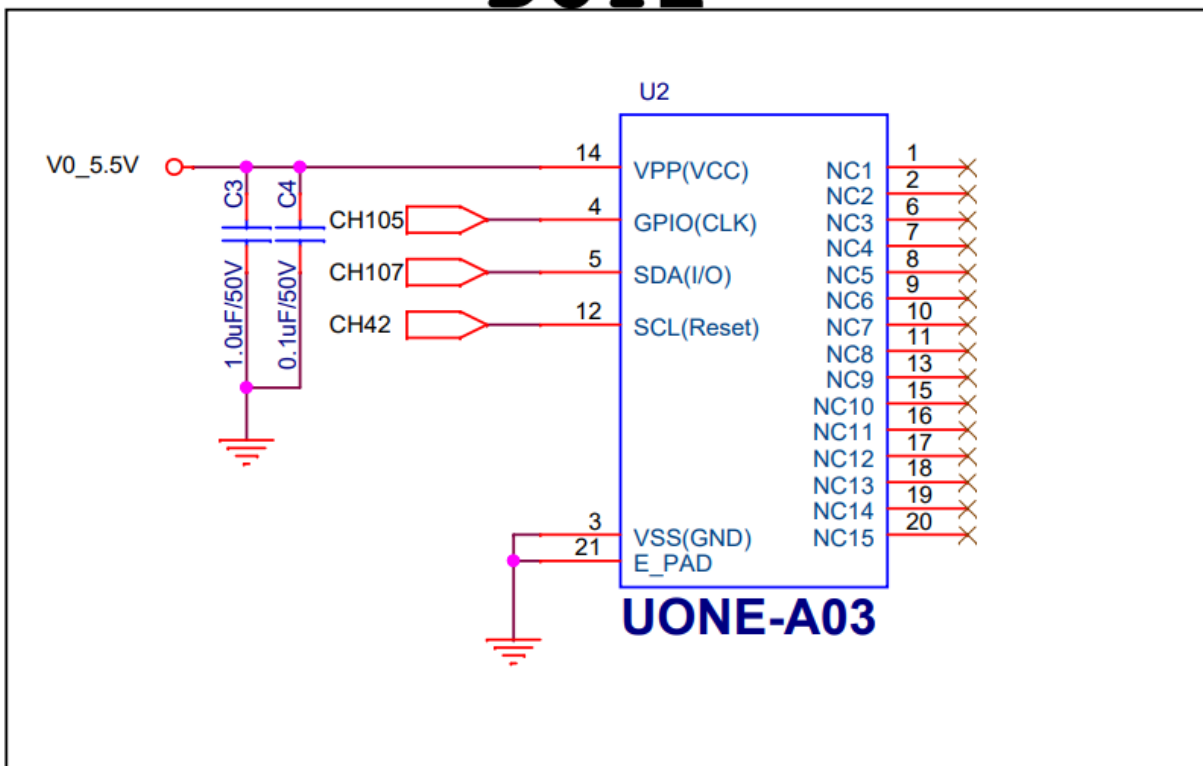
Product	:	UONE-A03
Package Type	:	20QFN
Board Type	:	HTOL Socket Board
Board S/N	:	BT22120A1
Material Code	:	BS3755
Quantity	:	40 ea

4. HTOL Board Circuit

DUT1



DUT2



5. HTOL Power Supply Waveform

V0 : 5.5V



6. HTOL Input Signal Waveform

CLK



7. HTOL Output Signal Waveform

SCL



SDA



8. HTOL Set-up Check Results

Step	Power	Voltage	Current	O.C.P	Check	Temp
1	V0	5.5 V	0 .042A	2A	OK	Room Temp
Remark	3ea loading					

	V0	V1	V2	V3	V4	
SLOT 3	5.500	0.000	0.000	0.000	0.000	Set Voltage
	5.497	0.000	0.001	-0.002	-0.001	Read Voltage
	0.042	0.001	0.002	0.001	0.001	Read Current



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